

Companion LXXXIX: Hierarchical Bayesian RDCC Inference with Multi-Probe Priors and Transport-Regularized Posteriors

Michael Lehmann

`mi.lehmann@gmx.de`

April 2026

Abstract

The Relaxation-Driven Cyclic Cosmology (RDCC) introduces the relaxation parameter α , which modifies the scale-dependent gravitational potential and induces observable signatures in weak-lensing convergence fields. Previous Companions developed transport-based topological statistics, multi-probe likelihoods, and emulator-accelerated inference. In this work, we construct a hierarchical Bayesian framework for RDCC inference that incorporates multi-probe priors, transport-regularized posteriors, and emulator uncertainty. The hierarchical structure enables principled combination of topological, morphological, and two-point observables, while transport regularization ensures geometric consistency across probes. This Companion provides the final statistical layer required for robust, scalable, and fully Bayesian RDCC parameter estimation.

1 Introduction

Weak-lensing surveys provide multiple complementary observables, including:

- topological transport statistics,
- Minkowski functionals,
- shear power spectra,
- neural joint likelihoods (Companion LXXXVI),
- emulator-accelerated predictions (Companion LXXXVIII).

A hierarchical Bayesian framework is required to:

- combine these probes coherently,
- propagate emulator uncertainty,
- incorporate survey systematics,
- regularize posteriors using geometric transport constraints.

This Companion constructs such a framework.

2 Hierarchical Model Structure

We define the hierarchical model:

$$\alpha \sim p(\alpha),$$

$$\theta_{\text{emu}} \sim p(\theta_{\text{emu}}),$$

$$\mathbf{s} \sim p(\mathbf{s} \mid \alpha, \theta_{\text{emu}}),$$

where:

- α is the RDCC relaxation parameter,
- θ_{emu} are emulator parameters,
- $\mathbf{s}$ is the multi-probe summary vector.

The posterior is:

$$p(\alpha, \theta_{\text{emu}} \mid \mathbf{s}_{\text{obs}}) \propto p(\mathbf{s}_{\text{obs}} \mid \alpha, \theta_{\text{emu}})p(\theta_{\text{emu}})p(\alpha).$$

3 Multi-Probe Priors

We define a prior that incorporates:

- theoretical RDCC constraints,
- emulator smoothness priors,
- cross-probe consistency priors.

The multi-probe prior is:

$$p(\alpha) \propto \exp \left[-\lambda_{\text{th}} \left(\frac{\partial^2 \mathcal{T}_\ell}{\partial \alpha^2} \right)^2 \right],$$

which penalizes unphysical curvature in transport statistics.

4 Transport-Regularized Posteriors

Transport regularization enforces geometric consistency across probes:

$$\mathcal{R}_{\text{trans}} = \sum_{\ell} W_2^2(S_{\theta}(\alpha), S_{\theta}(\alpha + \delta)),$$

where S_{θ} is the neural transport surrogate.

The posterior becomes:

$$p(\alpha \mid \mathbf{s}) \propto \mathcal{L}(\mathbf{s} \mid \alpha) \exp[-\lambda_{\text{trans}} \mathcal{R}_{\text{trans}}].$$

5 Inference and Sampling

The hierarchical posterior is sampled using:

- Hamiltonian Monte Carlo,
- No-U-Turn Sampler (NUTS),
- Variational inference,
- Neural posterior estimation.

Gradients are computed via automatic differentiation through:

- the neural joint likelihood (Companion LXXXVI),
- emulator flows (Companion LXXXVIII),
- transport regularization terms.

6 Conclusion

We introduced a hierarchical Bayesian framework for RDCC inference that combines multi-probe priors, emulator uncertainty, and transport-regularized posteriors. This framework unifies topology, morphology, and two-point statistics into a single coherent Bayesian model and provides the final statistical layer required for robust RDCC parameter estimation in Euclid-like surveys.

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